

SIMARPREET SINGH

Mukerian, Punjab, India | +917719668814 | simarpreetsaini96@gmail.com | [Linkedin](#) | [Github](#)

EDUCATION

I.K. Gujral Punjab Technical University, Kapurthala

2022 - 2026

B. Tech in Computer Science and Engineering

9.00 SGPA

EXPERIENCE

Ryazio Technologies LLP

Remote

Industrial Training

June 2024 – July 2024

- Built and optimized **full-stack** applications, focusing on performance and scalability.
- Enhanced skills in **RESTful API** development, authentication, and database management.
- Worked on **real-world** client projects, improving code quality and user experience.
- Strengthened problem-solving and industry-specific development practices.

Novem Controls

Remote

Web Development Internship

June 2024 – July 2024

- Developed and maintained full-stack web applications using the **MERN** stack.
- Designed responsive, user-friendly interfaces with **React.js** and **Tailwind CSS**.
- Implemented backend **APIs** with **Node.js** and **Express**, integrating **MongoDB** for scalable data storage.
- Collaborated with team members to deploy projects and resolve production-level issues.
- Gained hands-on experience in live project development and deployment workflows.

PROJECTS

BRSLMS – Full-Stack Learning Management System | *MERN Stack*

- Built a full-stack Learning Management System using React, Node.js, and Express to manage students, teachers, and academic data.
- Developed RESTful APIs for attendance tracking, assignments, and academic performance management.
- Implemented role-based authentication and authorization for admin and faculty users.
- Created an admin dashboard for managing users, records, and academic workflows.

Quantum Key Simulation – Satellite-Based Secure Key Distribution | *Python*

- Built a Python-based **Quantum Key Simulation (QKS)** using the BB84 protocol with a Tkinter GUI.
- Designed **interactive visualizations** for photon transmission, key reconciliation, and error detection.
- Implemented **QBER analysis** to detect eavesdropping and trigger secure key regeneration.
- Developed a **multi-tab GUI** showcasing simulation results, communication visualization, and system logs.

TECHNICAL SKILLS

Languages: JavaScript, Python, SQL, C++

Frontend: HTML, CSS, Tailwind CSS, Bootstrap, React.js

Backend: Node.js, Express.js, JWT

Database: MongoDB, MySQL, PostgreSQL, Firebase

Tools: VS Code, Git, GitHub, Firebase, RESTful APIs

Concepts: OOP, DSA

CERTIFICATES

Industrial Training certification – Gained practical exposure to coding standards, scalable solutions, and industry workflows.

Internship certification– Enhanced problem-solving and project execution abilities in a professional environment.